

Linear Search.

Ques-2

1) $m=5$

$arr = [1201, 1305, 1450, 1408, 1502]$

Key: 1450

	$arr[i]$	Key	Comparison	Result
$i=0$	1201	1450	$1201 == 1450$	False
$i=1$	1305	1450	$1305 == 1450$	False
$i=2$	1450	1450	$1450 == 1450$	True

If $(arr[i] == Key)$

↳ Fraud Transaction found.

Teacher's Signature.....

Linear Search

```
#include <iostream>
using namespace std;
int main() {
```

```
    int n;
    cin >> n;
    int arr[n];
    for(int i=0; i<n; i++) {
        cin >> arr[i];
    }
    int key;
    cin >> key;
    for(int i=0; i<n; i++) {
        if (arr[i] == key) {
```

```
            cout << "Found Transaction found";
            return 0;
        }
```

```
    }
    cout << "Transaction Not found";
    return 0;
}
```

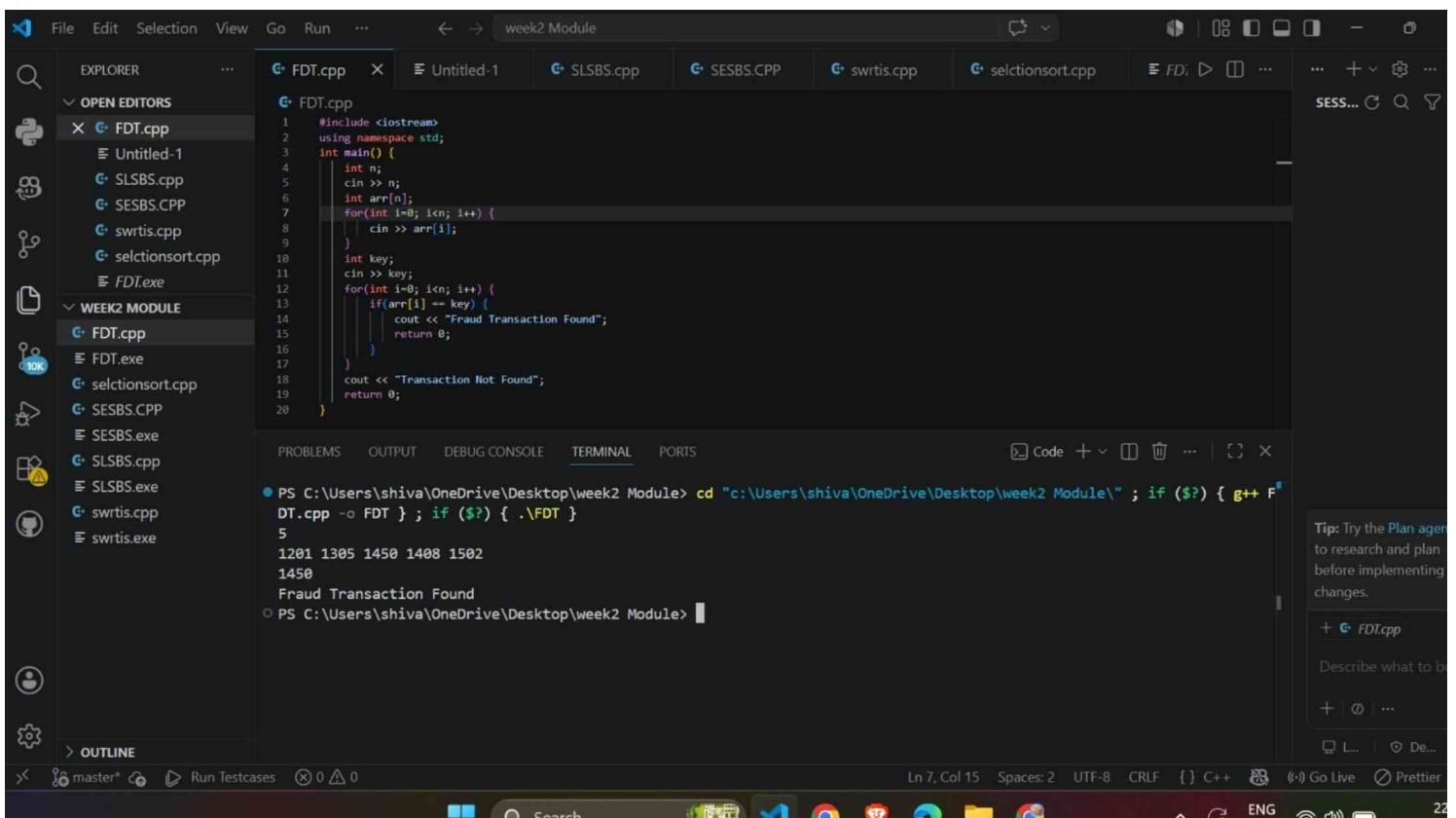
#Output 5

1201 1805 1450 1408 1502

Key: 1450

Found Transaction found

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Binary Search

दिनांक
पृष्ठ

i	arr[i]	Array
0	100	[100]
1	200	[100, 200]
2	300	[100, 200, 300]
3	400	[100, 200, 300, 400]
4	500	[100, 200, 300, 400, 500]

arr = [100, 200, 300, 400, 500]

ii) Target = 300

iii) Searching loop.

for(int i=0; i<n; i++){
if (arr[i] == target)

→ i=0

arr[0] = 100

target = 300

100 == 300? No

→ i++

→ i=1

arr[1] = 200

200 == 300? No

i++

→ i=2

300 == 300

Example
Log found

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Binary search

```

2) #include <iostream>
using namespace std;
int main() {
    int n;
    cin >> n;
    int arr[n];
    for(int i=0; i<n; i++) {
        cin >> arr[i];
    }
    int target;
    cin >> target;
    for(int i=0; i<n; i++) {
        if (arr[i] == target) {
            cout << "Log found";
            return 0;
        }
    }
    cout << "Log not found";
    return 0;
}

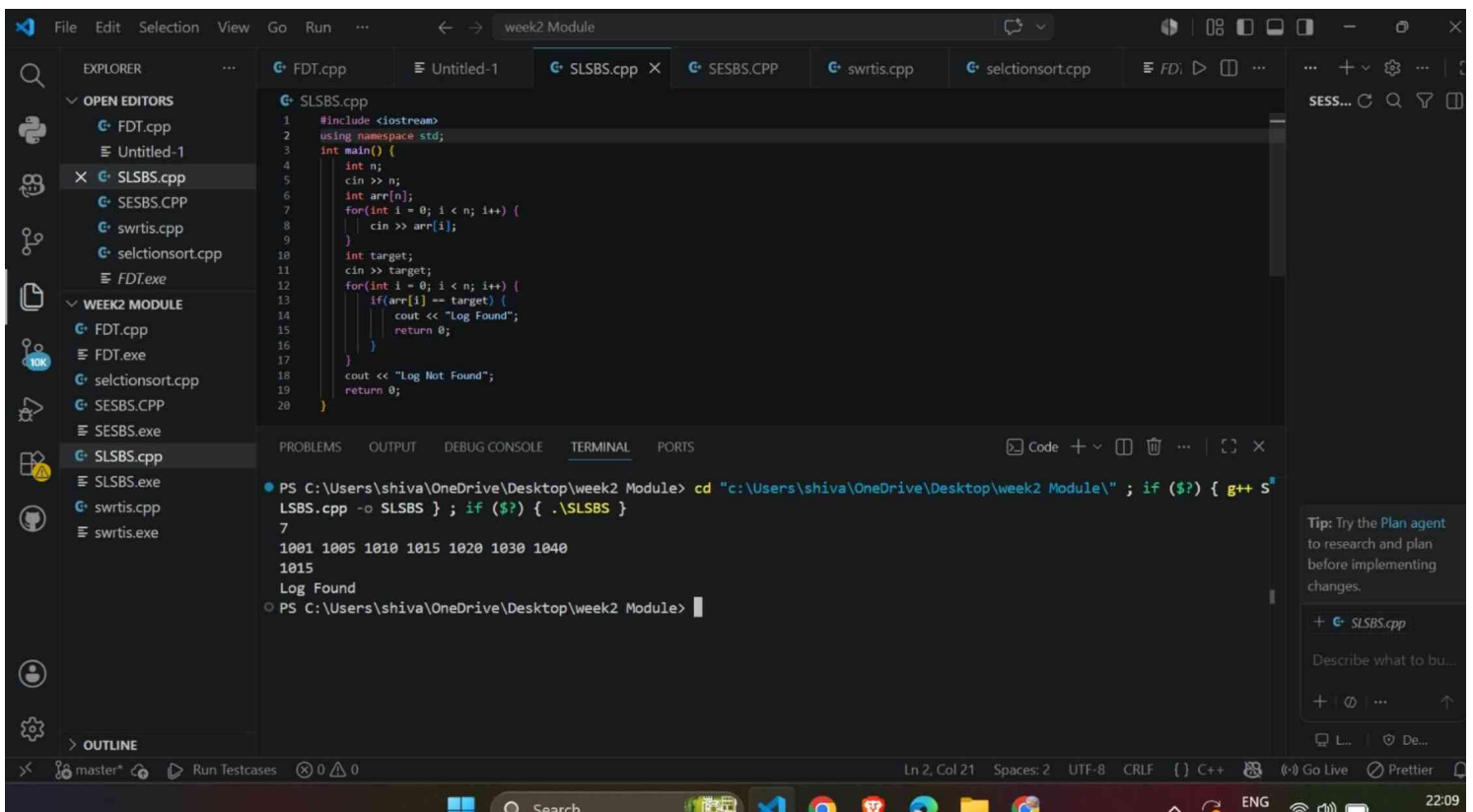
```

Input
~~#input~~ n=7

int arr[n] = 1001 1005 1010 1015 1020 1030 1040

#output: log found.

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Bubble sort

input = 5

arr = [45000, 32000, 55000, 28000, 60000]

(A) for (int i=0; i<n-1; i++)

ii) i=0
for (int j=0; j<n-i-1; j++)

j < 4

iii) arr[0] > arr[j]
j=0
true,
swap

(B) i=1

arr
j < 3
j=0

2) 32000 > 45000

No swap.

[32000, 45000, 55000, 28000, 60000]

j=1

3) 45000 > 28000

Swap.

iv) j=1
45000 > 55000
No swap.

[32000, 28000, 45000, 55000, 60000]

v) j=2

55000 > 28000

swap

[32000, 45000, 28000, 55000, 60000]

vi) j=2

No swap.

(C) i=2

j < 2

vii) j=3

No swap.

final output
[28000, 32000, 45000, 55000, 60000]

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Q 3. Bubble sort

```
#include <iostream>
using namespace std;

int main() {
    int n;
    cin >> n;
    int arr[n];
    for (int i = 0; i < n; i++) {
        cin >> arr[i];
    }

    for (int i = 0; i < n - 1; i++) {
        for (int j = 0; j < n - i - 1; j++) {
            if (arr[j] > arr[j + 1]) {
                swap(arr[j], arr[j + 1]);
            }
        }
    }

    for (int i = 0; i < n; i++) {
        cout << arr[i] << " ";
    }

    return 0;
}
```

Input

n = 5

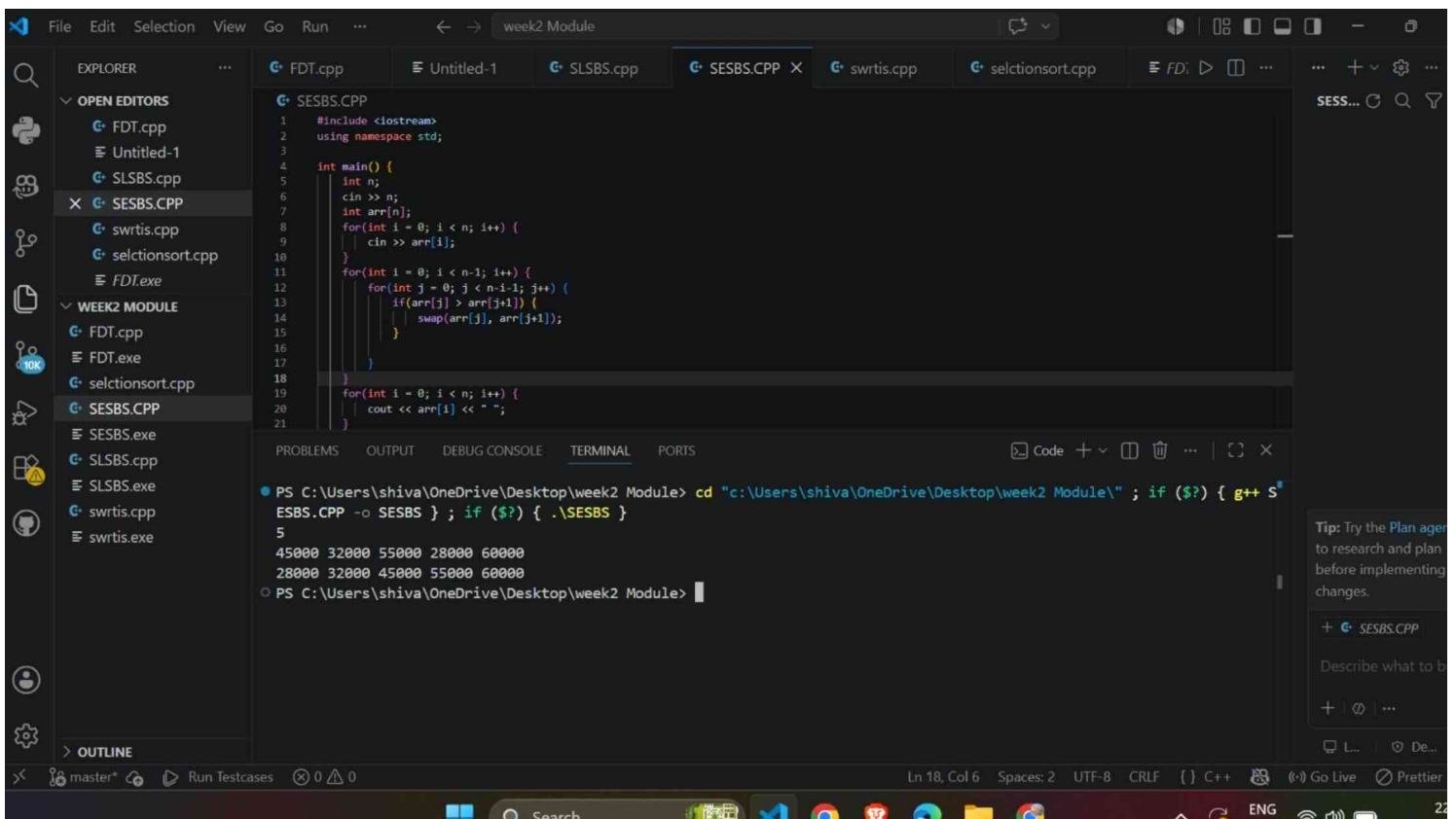
int arr[5]

45000 32000 55000 28000 67000

Output

28000 32000 45000 55000 67000

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Insertion sort

$n=5$

$arr = [250, 120, 320, 180, 100]$

A) $i=1$

$key = arr[i]$

$key = 120$

$j = i-1$

$j = 0$

$[250, 120, 320, 180, 100]$

$\hookrightarrow 0 \geq 0$ $250 > 120$

$j \geq 0$ $arr[j] > key$

move ~~8~~ one step right

$arr[j+1] = arr[j]$

$[250, 250, 320, 180, 100]$

$j = j-1$

$arr[j+1] = key$

$[120, 250, 320, 180, 100]$

$i=2$

$key = arr[2]$

$key = 320$

No $500 > 320$

$i=3$

$key = arr[3]$

$key = 180$

$j=2$

$320 > 180$

$[120, 250, 180, 320, 100]$

$[100, 120, 180, 250, 320]$

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Insertion Sort

```

4) #include <iostream>
using namespace std;
int main()

{
    int n;
    cin >> n;
    int arr[n];

    for(int i=0; i<n; i++)

    {
        cin >> arr[i];
    }

    for (int i=1; i<n; i++)
    {
        int key = arr[i];
        int j = i-1;

        while(j >= 0 & arr[j] > key)

        {
            arr[j+1] = arr[j];
            j--;
        }
        arr[j+1] = key;
    }

    for (int i=0; i<n; i++)

    {
        cout << arr[i] << " ";
    }

    return 0;
}

```

```

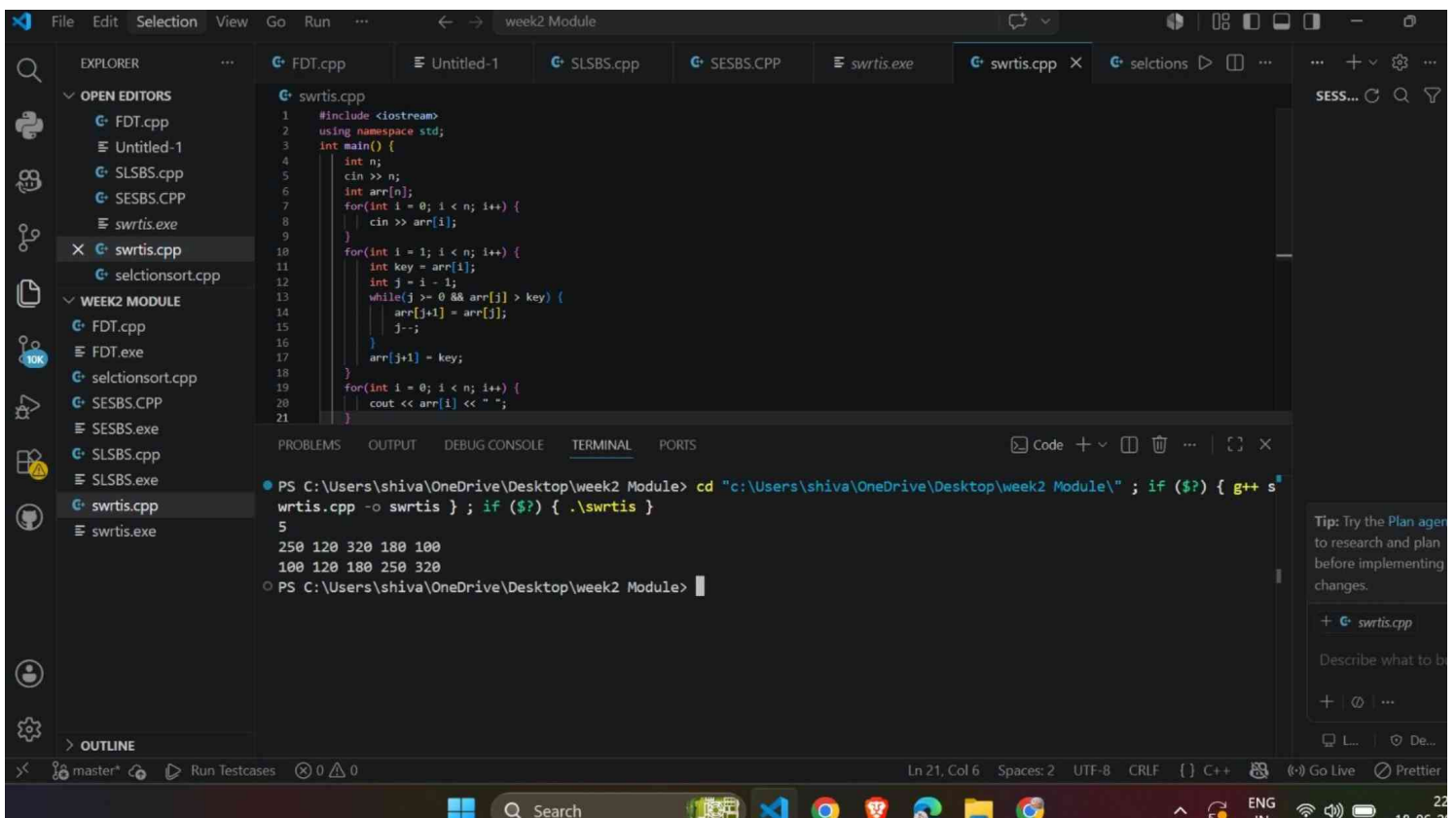
input n=5
int arr[5]

250 120 320 180 100

#output
100 120 180 250 320

```

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Selection Sort

Step 1 $\hookrightarrow i=0$

$\text{minIndex} = 0$

$\text{arr} = [9, 2, 8, 1, 5]$

$\hookrightarrow j=1$

$\text{arr}[j] < \text{arr}[\text{minIndex}]$

$2 < 9$

$\text{minIndex} = j$

$\hookrightarrow j=2$

$8 < 2$

false

$\hookrightarrow j=3$

$1 < 2$

false update

$\text{minIndex} = 3$

$\hookrightarrow j=4$

$5 < 1$

false

Now, swap ($\text{arr}[0], \text{arr}[3]$)

$[1, 2, 8, 9, 5]$

Step 2

outer loop 2

$\hookrightarrow i=1$

$\text{minIndex} = j$

Array

$[1, 2, 8, 9, 5]$

$\hookrightarrow$ Inner loop

$j=2$

$8 < 2$

false

$\hookrightarrow j=3$

$9 < 2$

$\hookrightarrow j=4$

$5 < 2$

false.

No swap

Step 3

$i=2$

$\text{minIndex} = 2$

$j=3$

$8 < 8$ false.

$\hookrightarrow j=4$

$5 < 1$ update

$\text{minIndex} = j$

$[1, 2, 5, 9, 8]$

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~~outer loop 4~~

outer loop 4

$i = 3$

$\text{minIndex} = 3$

$[1, 2, 5, 9, 8]$

Inner loop

$j = 4$

$\hookrightarrow j < 9$
true

$\text{minIndex} = 4$

Swap $\text{arr}[3]$ and $\text{arr}[4]$

Array

$[1, 2, 5, 8, 9]$

Final ans

$[1, 2, 5, 8, 9]$

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Selection Sort

```
#include <iostream>
using namespace std;

int main()
{
    int n;
    cin >> n;
    int arr[n];

    for (int i = 0; i < n; i++)
        cin >> arr[i];

    for (int i = 0; i < n - 1; i++)
    {
        int minIndex = i;

        for (int j = i + 1; j < n; j++)
        {
            if (arr[j] < arr[minIndex])
            {
                minIndex = j;
            }
        }
        swap(arr[i], arr[minIndex]);
    }

    for (int i = 0; i < n; i++)
        cout << arr[i] << " ";

    return 0;
}
```

input
n = 5
int arr[5]
9 2 8 1 5
output
1 2 5 8 9

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